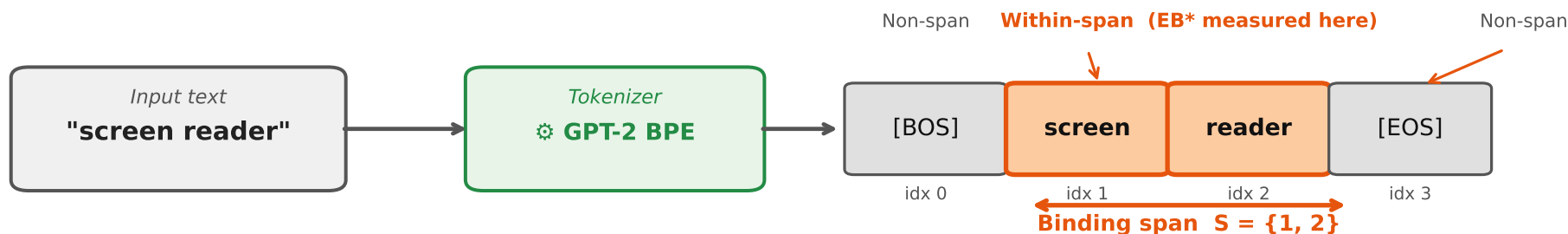
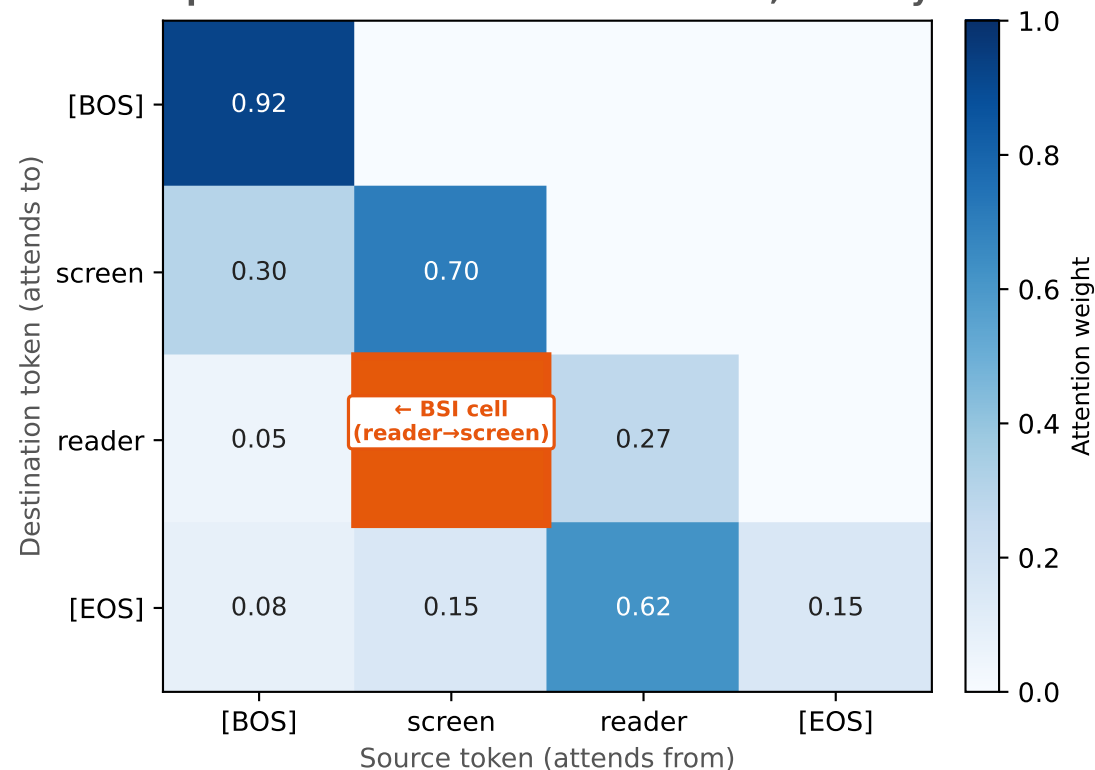


Attention-Head Binding (EB*): From Token Span to Mechanistic Signal

Step 1: Multi-Token Term → Attention Span



Step 2: Attention Matrix — One Head, One Layer



Step 3: EB* Calculation

$$\text{BSI}(T, \ell, h) = \text{mean attention}(\text{dest} \rightarrow \text{src})$$

for $\text{dest}, \text{src} \in S, \text{dest} > \text{src}$

(average later-to-earlier within-span attention for head h at layer ℓ)

$$\text{EB}(T, \ell) = \max_h \text{BSI}(T, \ell, h) - \text{mean}_h \text{BSI}(T, \ell, h)$$

(prominence of the single most binding head vs. layer average)

$$\text{EB}^*(T) = \max_{\ell} \text{EB}(T, \ell)$$

(maximum binding prominence across all layers)

Why EB* is lightweight

- ✓ Forward-pass only — no training required
- ✓ Reads 2-4 attention cells, not full seq-len matrix
- ✓ ~1 MB per pass vs. TBs for full attention at 405B
- ✓ Compatible with GQA (per query-head)